

Mahatru Miryala

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Experience

Quality Engineer I - Arthrex, Santa Barbara

Oct 2025 - Current

- Supported sustaining engineering for Electrosurgical Generators, Camera Consoles, and Touch Screen Components across Receiving Inspection, Manufacturing, and Global Service and Repair.
- Supported root cause analysis of intermittent console fan failures, identifying insufficient solder on a board grounding pin and supporting X-ray inspection of console inventory.
- Supported engineering assessment of fan-off thermal performance and supplier communication on corrective process changes; the study found no overheating risk or thermal shutdown under evaluated conditions.
- Authored work instructions and supported internal/external audits and international product-line setup in China.

Quality Engineer Intern - Arthrex, Santa Barbara

June 2025 - Oct 2025

- Introduced **Agile** methodologies and worked with IT, developing business justifications to evaluate **Jira** and **Workfront** and roll out to support team workflows and improve project timelines. Also worked with Quality Control to inspect incoming parts.

Undergraduate Research Assistant - Coskun Lab, Atlanta

Jan 2025 - May 2025

- Supported research into computer vision detection and visualization of disease progression in kidney tissues in both kidney tissue handling and imaging and data labeling for the ML model.

Test Engineer Intern - Spectaire, Boston

June 2024 - Aug 2024

- Developed Python test software for portable mass spectrometers and SQL/PostgreSQL infrastructure to capture test and field data; standardized the application for manufacturer use during assembly.
- Supported product assembly and delivered and set up units at customer sites.

Undergraduate Research Assistant - Robomed Lab, Atlanta

Aug 2023 - Dec 2023

- Research and development of robotically steerable guidewires for endovascular and cardiovascular applications.
- Improved image distortion algorithm runtime by 92%, created and soldered PCBs for mechanism assembly for animal testing.

Undergraduate Research Assistant - WEAR Lab, Orlando

Feb 2022 - Aug 2022

- Spearheaded research leveraging object recognition, hand-tracking, and LiDAR to empower mobility in blind people.
- Used convolutional neural networks and hidden Markov models to translate gestures into voice messages for research intended to support Army units in situations requiring silence.

Projects

Astroject, A NASA Senior Design Project - Project Manager

Aug 2024 - Dec 2024

- Coordinated team resources and deadlines and contributed to system design for a NASA-sponsored medication-delivery concept intended to operate through a spacesuit while preserving suit integrity.

[NASA L'SPACE Mission Concept Academy](#) - Astrobiologist

Jan 2022 - April 2022

- Contributed logistics, systems concepts, and protocols to a 15-person team's lunar oxygen mission proposal; co-developed a 78-page preliminary design review (PDR) report and presented to a review board.

[Parkinson's Detection Device](#) - Lead Developer

Jan 2023 - May 2023

- Led a six-person team developing an Arduino-based gait-monitoring prototype that collected accelerometer and gyroscope data for machine-learning analysis.
- Created website, databases via **AWS**, **Arduino software**, and AI model with **TensorFlow**.

Education & Extracurriculars

Purdue University: West Lafayette, Indiana

August 2026 - Present

- Space Systems Engineering MS (remote)

Georgia Institute of Technology: Atlanta, GA

Class of 2025

- Biomedical Engineering BS, Industrial Design Minor
- Graduated in the Honors Program With Research Distinction

BSA Troop 125 - Eagle Scout

2014 - Present

- Led 30+ volunteers over 180+ service hours to build a dog agility course for a veteran-support dog program; raised \$1,100.

Skills

Engineering: Verification and Validation (V&V), Root Cause Analysis, DFMEA, PFMEA, GD&T, SolidWorks, Fusion360

Programming: Java, C, Python, SQL, PostgreSQL, MATLAB, Arduino, TensorFlow, AWS